

Writing Small Numbers in Scientific Notation

Teacher's Edition

The Period

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Must Express very small quantities in scientific notation and read them back; determine how many times larger or smaller one number is than another.

Must not Coefficients are at least 1 and less than 10. A negative exponent never makes the value negative, and operations in scientific notation are Unit 4.

Target I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.

MTR MTR.5.1 — Notes I · **MTR.6.1** — Board 7 · **MTR.7.1** — Board 9 and the bank

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	13 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. Small number, negative exponent. The minus is about size; it never makes the number negative.
2. The exponent counts the places the decimal point moves, not the zeros you can see.
3. Compare the exponents first. The coefficients only break a tie.

Slide by Slide

1. **Writing Small Numbers in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q2 and Q4 are the two halves of today: 10^{-3} is a thousandth, and 0.007 is 7 thousandths, so 0.007 is 7×10^{-3} .
3. **Warm-Up — Answers.** Reveal. 1. 4.3×10^6 2. $\frac{1}{1,000}$ 3. 9 and 10 4. $\frac{7}{1,000}$
4. **Notes — Copy the table.** The exponent tells you which decimal place the 1 is in. Not how many zeros.
5. **Notes.** Find the first nonzero digit. Its place is the exponent. Everything else is the coefficient.
6. **Notes.** The larger exponent wins. -4 is larger than -7 , so 10^{-4} is the bigger power.
7. **Example 1 — A quantity in words, to scientific notation.** SPOKEN ONLY.
8. **Example 1 — Worked.** 1,332 millionths is $1,332 \div 1,000,000 = 0.001332$.
9. **Your Turn — Same steps, your numbers. Boards up when done.** The book's human hair.
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — How many times larger?** SPOKEN ONLY.
12. **Example 2 — Worked.** $7.2 \div 1.8 = 4$ and $10^{-4} \div 10^{-7} = 10^3$ (-4 minus -7 is 3).
13. **Your Turn — Same steps, your numbers. Boards up when done.** The book's practice item.
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** $10^{-4} \rightarrow$ **0.0001** 0.00001 — four zeros written, then the 1 (five places) [8.NSO.1.4: the exponent counts places]; $-10,000$ — negative exponent read as a negative value [B1G-M 8.NSO.1.3 misconception]
- 17-18. **Board 2.** Write 0.003 in scientific notation. $\rightarrow$ **3×10^{-3}** 3×10^3 — the sign was dropped: that is 3,000 [8.NSO.1.4: numbers less than 1 need a negative exponent]; 3×10^{-2} — counted the zeros instead of the place
- 19-20. **Board 3.** Write 2.5×10^{-4} in standard form. $\rightarrow$ **0.00025** 0.000025 — point moved five places [8.NSO.1.4: 10^{-4} moves the point four places]; 25,000 — point moved right instead of left [negative exponent read as a large number]
- 21-22. **Board 4.** A machine's cutting tolerance is 0.0000988 inch. The part it cuts is 4 inches long. Write the tolerance in scientific notation. $\rightarrow$ **9.88×10^{-5}** The 4-inch length of the part is not needed. 9.88×10^{-4} — counted the zeros [8.NSO.1.4: exponent = places moved]; 988×10^{-7} — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception]; anything built from 4 — the unneeded figure was used
- 23-24. **Board 5.** Which is 0.00081 written in scientific notation? $\rightarrow$ **D — 8.1×10^{-4}** (A) counted the three zeros instead of the four places [8.NSO.1.4: exponent = places moved]; (B) the sign was dropped: 8.1×10^4 is 81,000 [8.NSO.1.4: numbers less than 1 need a negative exponent]; (C) equal in value, but 81 is not less than 10 [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.
- 25-26. **Board 6.** How many times larger is 6×10^{-3} than 2×10^{-5} ? $\rightarrow$ **A — 300** (B) 10^2 read as 10 [B1G-M 8.NSO.1.4 misconception: comparing two numbers in

scientific notation incorrectly]; (C) coefficients only — the powers were ignored [8.NSO.1.4: both factors are compared]; (D) the exponents were subtracted the wrong way, $-5 - (-3) = -2$ [8.NSO.1.3: quotient law, top minus bottom] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

27-28. Board 7. Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ? → **3.9×10^{-5}**
 7.4×10^{-8} — coefficients compared first, or -8 read as larger than -5 [8.NSO.1.4: compare the powers of 10 first, and $-8 < -5$]

29-30. Board 8. A computer displays a number as $4.2\text{E}-6$. Write the number in standard form. → **0.0000042** $-4,200,000$ — the exponent's sign read as the number's sign [B1G-M 8.NSO.1.4 misconception: confusing the exponent with the value of the number]; 0.00000042 — point moved seven places

31-32. Board 9. Fishing line A is 2×10^{-2} inch thick. Fishing line B is 6×10^{-2} inch thick. Which is thicker, and how many times thicker? Give a possible thickness between the two. → **B; 3 times; e.g. 4×10^{-2} in.** ' 2×10^{-2} is thicker' — 10^{-2} read as making a smaller coefficient bigger; ' 4×10^{-1} in between' — power ignored in the in-between value [8.NSO.1.4: both factors matter]

33. Independent Set — Six questions. On your own, in silence. Hand out the Independent Set.

34. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 0.00001 — four zeros written, then the 1 (five places) [8.NSO.1.4: the exponent counts places]

Board 2. 3×10^3 — the sign was dropped: that is 3,000 [8.NSO.1.4: numbers less than 1 need a negative exponent]

Board 3. 0.000025 — point moved five places [8.NSO.1.4: 10^{-4} moves the point four places]

Board 4. 9.88×10^{-4} — counted the zeros [8.NSO.1.4: exponent = places moved]

Board 5. counted the three zeros instead of the four places [8.NSO.1.4: exponent = places moved]

Board 6. 10^2 read as 10 [B1G-M 8.NSO.1.4 misconception: comparing two numbers in scientific notation incorrectly]

Board 7. 7.4×10^{-8} — coefficients compared first, or -8 read as larger than -5 [8.NSO.1.4: compare the powers of 10 first, and $-8 < -5$]

Board 8. $-4,200,000$ — the exponent's sign read as the number's sign [B1G-M 8.NSO.1.4 misconception: confusing the exponent with the value of the number]

Board 9. ' 2×10^{-2} is thicker' — 10^{-2} read as making a smaller coefficient bigger

Materials Whiteboards and markers. Calculators are allowed on everything; entering 4.2×10^{-6} and reading the display back as $4.2\text{E}-6$ is worth thirty seconds at whiteboard 8.